

Theory of Computation (TOC)

GATE Computer Science & IT Previous Year Questions (2016 – 2025) | Total Questions: 73

Question 1 of 73 | GATE 2025 Set 1 (CS1) | Q.19 (Computer Science & IT)

Q.19

Consider the following context-free grammar G , where S , A , and B are the variables (non-terminals), a and b are the terminal symbols, S is the start variable, and the rules of G are described as:

$S \rightarrow aaB \mid Abb$

$A \rightarrow a \mid aA$

$B \rightarrow b \mid bB$

Which ONE of the languages $L(G)$ is accepted by G ?

- (A) $L(G) = \{a^2bn \mid n \geq 1\} \cup \{anb^2 \mid n \geq 1\}$
- (B) $L(G) = \{anb^2n \mid n \geq 1\} \cup \{a^2nb^n \mid n \geq 1\}$
- (C) $L(G) = \{anbn \mid n \geq 1\}$
- (D) $L(G) = \{a^2nb^2n \mid n \geq 1\}$

Question 2 of 73 | GATE 2025 Set 1 (CS1) | Q.28 (Computer Science & IT)

Q.28

A regular language L is accepted by a non-deterministic finite automaton (NFA) with n states. Which of the following statement(s) is/are FALSE?

- (A) L may have an accepting NFA with $< n$ states.
- (B) L may have an accepting DFA with $< n$ states.
- (C) There exists a DFA with $\leq 2n$ states that accepts L .
- (D) Every DFA that accepts L has $> 2n$ states.

Question 3 of 73 | GATE 2025 Set 1 (CS1) | Q.30 (Computer Science & IT)

Q.30

Let S be the set of all ternary strings defined over the alphabet $\{a, b, c\}$. Consider all strings in S that contain at least one occurrence of two consecutive symbols, that is, "aa", "bb" or "cc". The number of such strings of length 5 that are possible is _____. (Answer in integer)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 4 of 73 | GATE 2025 Set 1 (CS1) | Q.44 (Computer Science & IT)

Q.44

Consider the following two languages over the alphabet $\{a, b\}$:

$$L1 = \{ \alpha\beta\alpha \mid \alpha \in \{a, b\}^+ \text{ AND } \beta \in \{a, b\}^+ \}$$

$$L2 = \{ \alpha\beta\alpha \mid \alpha \in \{a\}^+ \text{ AND } \beta \in \{a, b\}^+ \}$$

Which ONE of the following statements is CORRECT?

- (A) Both $L1$ and $L2$ are regular languages.
- (B) $L1$ is a regular language but $L2$ is not a regular language.
- (C) $L1$ is not a regular language but $L2$ is a regular language.
- (D) Neither $L1$ nor $L2$ is a regular language.

Question 5 of 73 | GATE 2025 Set 1 (CS1) | Q.45 (Computer Science & IT)

Q.45

Consider the following two languages over the alphabet $\{a, b, c\}$, where m and n are natural numbers.

$$L1 = \{ ambmcm+n \mid m, n \geq 1 \}$$

$$L2 = \{ ambncm+n \mid m, n \geq 1 \}$$

Which ONE of the following statements is CORRECT?

- (A) Both $L1$ and $L2$ are context-free languages.
- (B) $L1$ is a context-free language but $L2$ is not a context-free language.
- (C) $L1$ is not a context-free language but $L2$ is a context-free language.
- (D) Neither $L1$ nor $L2$ are context-free languages.

Question 6 of 73 | GATE 2025 Set 1 (CS1) | Q.50 (Computer Science & IT)

Q.50

Consider the following deterministic finite automaton (DFA) defined over the alphabet, $\Sigma = \{a, b\}$. Identify which of the following language(s) is/are accepted by the given DFA.

- (A) The set of all strings containing an even number of b 's.
- (B) The set of all strings containing the pattern bab .
- (C) The set of all strings ending with the pattern bab .
- (D) The set of all strings not containing the pattern aba .

Question 7 of 73 | GATE 2025 Set 2 (CS2) | Q.24 (Computer Science & IT)

Q.24

Which ONE of the following languages is accepted by a deterministic pushdown automaton?

- (A) Any regular language.
- (B) Any context-free language.
- (C) Any language accepted by a non-deterministic pushdown automaton.
- (D) Any decidable language.

Question 8 of 73 | GATE 2025 Set 2 (CS2) | Q.25 (Computer Science & IT)

Q.25

Let G_1, G_2 be Context Free Grammars (CFGs) and R be a regular expression. For a grammar G , let $L(G)$ denote the language generated by G .

Which ONE among the following questions is decidable?

- (A) Is $L(G_1) = L(G_2)$?
- (B) Is $L(G_1) \cap L(G_2) = \emptyset$?
- (C) Is $L(G_1) = L(R)$?
- (D) Is $L(G_1) = \emptyset$?

Question 9 of 73 | GATE 2025 Set 2 (CS2) | Q.30 (Computer Science & IT)

Q.30

Consider the two lists List I and List II given below:

For matching of items in List I with those in List II, which of the following option(s) is/are CORRECT?

List I

List II

- (i) Context free languages
- (a) Closed under union
- (ii) Recursive languages
- (b) Not closed under complementation
- (iii) Regular languages
- (c) Closed under intersection

- (A) (i) – (a), (ii) – (b), and (iii) – (c)
- (B) (i) – (b), (ii) – (a), and (iii) – (c)
- (C) (i) – (b), (ii) – (c), and (iii) – (a)
- (D) (i) – (a), (ii) – (c), and (iii) – (b)

Question 10 of 73 | GATE 2025 Set 2 (CS2) | Q.37 (Computer Science & IT)

Q.37

Let G be an edge-weighted undirected graph with positive edge weights. Suppose a

positive constant α is added to the weight of every edge.

Which ONE of the following statements is TRUE about the minimum spanning trees (MSTs) and shortest paths (SPs) in G before and after the edge weight update?

- (A) Every MST remains an MST, and every SP remains an SP.
- (B) MSTs need not remain MSTs, and every SP remains an SP.
- (C) Every MST remains an MST, and SPs need not remain SPs.
- (D) MSTs need not remain MSTs, and SPs need not remain SPs.

Question 11 of 73 | GATE 2025 Set 2 (CS2) | Q.40 (Computer Science & IT)

Q.40

Given a Context-Free Grammar G as follows:

$S \rightarrow Aa \mid bAc \mid dc \mid bda$

$A \rightarrow d$

Which ONE of the following statements is TRUE?

- (A) G is neither LALR(1) nor SLR(1)
- (B) G is CLR(1), not LALR(1)
- (C) G is LALR(1), not SLR(1)
- (D) G is LALR(1), also SLR(1)

Question 12 of 73 | GATE 2025 Set 2 (CS2) | Q.51 (Computer Science & IT)

Q.51

Consider two grammars G_1 and G_2 with the production rules given below:

$G_1: S \rightarrow \text{if } E \text{ then } S \mid \text{if } E \text{ then } S \text{ else } S \mid a$

$E \rightarrow b$

$G_2: S \rightarrow \text{if } E \text{ then } S \mid M$

$M \rightarrow \text{if } E \text{ then } M \text{ else } S \mid c$

$E \rightarrow b$

where if, then, else, a , b , c are the terminals.

Which of the following option(s) is/are CORRECT?

- (A) G_1 is not LL(1) and G_2 is LL(1).
- (B) G_1 is LL(1) and G_2 is not LL(1).
- (C) G_1 and G_2 are not LL(1).
- (D) G_1 and G_2 are ambiguous.

Question 13 of 73 | GATE 2025 Set 2 (CS2) | Q.52 (Computer Science & IT)

Q.52

Let $\Sigma = \{a, b, c\}$. For $x \in \Sigma^*$, and $\alpha \in \Sigma$, let $\# \alpha(x)$ denote the number of occurrences of α in x .

Which one or more of the following option(s) define(s) regular language(s)?

- (A) $\{ambn \mid m, n \geq 0\}$
- (B) $\{a, b\}^* \cap \{ambncm - n \mid m \geq n \geq 0\}$
- (C) $\{w \mid w \in \{a, b\}^*, \#a(w) \equiv 2 \pmod{7}, \text{ and } \#b(w) \equiv 3 \pmod{9}\}$
- (D) $\{w \mid w \in \{a, b\}^*, \#a(w) \equiv 2 \pmod{7}, \text{ and } \#a(w) = \#b(w)\}$

Question 14 of 73 | GATE 2025 Set 2 (CS2) | Q.60 (Computer Science & IT)

Q.60

Let $\Sigma = \{1, 2, 3, 4\}$. For $x \in \Sigma^*$, let $\text{prod}(x)$ be the product of symbols in x modulo 7. We take $\text{prod}(\epsilon) = 1$, where ϵ is the null string.

For example, $\text{prod}(124) = (1 \times 2 \times 4) \bmod 7 = 1$.

Define $L = \{x \in \Sigma^* \mid \text{prod}(x) = 2\}$.

The number of states in a minimum state DFA for L is _____. (Answer in integer)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 15 of 73 | GATE 2024 Set 1 (CS1 S5) | Q.23 (Computer Science & IT)

Q.23

Let L_1, L_2 be two regular languages and L_3 a language which is not regular.

Which of the following statements is/are always TRUE?

- (A) $L_1 = L_2$ if and only if $L_1 \cap L_2^c = \emptyset$
- (B) $L_1 \cup L_3$ is not regular
- (C) L_3^c is not regular
- (D) $L_1^c \cap L_2^c$ is regular

Question 16 of 73 | GATE 2024 Set 1 (CS1 S5) | Q.38 (Computer Science & IT)

Q.38

Consider the following grammar G , with S as the start symbol. The grammar G has three incomplete productions denoted by (1), (2), and (3).

$S \rightarrow daT \mid$ (1)

$T \rightarrow aS \mid bT \mid$ (2)

$R \rightarrow$ (3) $\mid \epsilon$

The set of terminals is $\{a, b, c, d, f\}$. The FIRST and FOLLOW sets of the different non-terminals are as follows.

$\text{FIRST}(S) = \{c, d, f\}$, $\text{FIRST}(T) = \{a, b, \epsilon\}$, $\text{FIRST}(R) = \{c, \epsilon\}$

$\text{FOLLOW}(S) = \text{FOLLOW}(T) = \{c, f, \$\}$, $\text{FOLLOW}(R) = \{f\}$

Which one of the following options CORRECTLY fills in the incomplete productions?

- (A) (1) $S \rightarrow Rf$ (2) $T \rightarrow \epsilon$ (3) $R \rightarrow cTR$
- (B) (1) $S \rightarrow fR$ (2) $T \rightarrow \epsilon$ (3) $R \rightarrow cTR$
- (C) (1) $S \rightarrow fR$ (2) $T \rightarrow cT$ (3) $R \rightarrow cR$
- (D) (1) $S \rightarrow Rf$ (2) $T \rightarrow cT$ (3) $R \rightarrow cR$

Question 17 of 73 | GATE 2024 Set 1 (CS1 S5) | Q.40 (Computer Science & IT)

Q.40

Consider the following two threads T1 and T2 that update two shared variables a and b. Assume that initially $a = b = 1$. Though context switching between threads can happen at any time, each statement of T1 or T2 is executed atomically without interruption.

T1 T2

$a = a + 1; b = 2 * b;$

$b = b + 1; a = 2 * a;$

Which one of the following options lists all the possible combinations of values of a and b after both T1 and T2 finish execution?

- (A) $(a = 4, b = 4); (a = 3, b = 3); (a = 4, b = 3)$
- (B) $(a = 3, b = 4); (a = 4, b = 3); (a = 3, b = 3)$
- (C) $(a = 4, b = 4); (a = 4, b = 3); (a = 3, b = 4)$
- (D) $(a = 2, b = 2); (a = 2, b = 3); (a = 3, b = 4)$

Question 18 of 73 | GATE 2024 Set 1 (CS1 S5) | Q.50 (Computer Science & IT)

Q.50

Consider the 5-state DFA M accepting the language $L(M) = (0 + 1)^*$ shown below. For any string $w \in (0 + 1)^*$ let $n_0(w)$ be the number of 0's in w and $n_1(w)$ be the number of 1's in w.

Which of the following statements is/are FALSE?

- (A) States 2 and 4 are distinguishable in M
- (B) States 3 and 4 are distinguishable in M
- (C) States 2 and 5 are distinguishable in M
- (D) Any string w with $n_0(w) = n_1(w)$ is in L(M)

Question 19 of 73 | GATE 2024 Set 1 (CS1 S5) | Q.59 (Computer Science & IT)

Q.59

Let $G = (V, \Sigma, S, P)$ be a context-free grammar in Chomsky Normal Form with $\Sigma = \{a, b, c\}$ and V containing 10 variable symbols including the start symbol S . The string $w = a30b30c30$ is derivable from S . The number of steps (application of rules) in the derivation $S \rightarrow^* w$ is _____

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[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 20 of 73 | GATE 2024 Set 1 (CS1 S5) | Q.61 (Computer Science & IT)

Q.61

Consider the following two regular expressions over the alphabet $\{0,1\}$:

 $r = 0^* + 1^*$ $s = 01^* + 10^*$

The total number of strings of length less than or equal to 5, which are neither in r nor in s , is _____

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 21 of 73 | GATE 2024 Set 2 (CS2 S6) | Q.22 (Computer Science & IT)

Q.22

Which one of the following regular expressions is equivalent to the language accepted by the DFA given below?

1

1

0

0

(A) $0^*1(0 + 10^*1)^*$ (B) $0^*(10^*11)^*00^*$ (C) $0^*1(010^*1)^*00^*$ (D) $0(1 + 0^*10^*1)^*00^*$ Organizing Institute: IISc, Bengaluru**Question 22 of 73 | GATE 2024 Set 2 (CS2 S6) | Q.29 (Computer Science & IT)**

Q.29

Which of the following statements is/are FALSE?

(A) An attribute grammar is a syntax-directed definition (SDD) in which the functions in the semantic rules have no side effects

(B) The attributes in a L-attributed definition cannot always be evaluated in a depth- first order

(C) Synthesized attributes can be evaluated by a bottom-up parser as the input is parsed

(D) All L-attributed definitions based on LR(1) grammar can be evaluated using a bottom-up parsing strategy

Question 23 of 73 | GATE 2024 Set 2 (CS2 S6) | Q.40 (Computer Science & IT)

Q.40

Consider the following context-free grammar where the start symbol is S and the set of terminals is {a,b,c,d}.

$S \rightarrow AaAb \mid \square BbBa$

$A \rightarrow cS \mid \square \epsilon$

$B \rightarrow dS \mid \square \epsilon$

The following is a partially-filled LL(1) parsing table.

a	
b	
c	
d	
\$	
S	
$S \rightarrow AaAb$	
$S \rightarrow BbBa$	
(1)	
(2)	
A	
$A \rightarrow \epsilon$	
(3)	
$A \rightarrow cS$	
B	
(4)	
$B \rightarrow \epsilon$	
$B \rightarrow dS$	

Which one of the following options represents the CORRECT combination for the numbered cells in the parsing table?

Note: In the options, "blank" denotes that the corresponding cell is empty.

- (A) (1) $S \rightarrow AaAb$ (2) $S \rightarrow BbBa$ (3) $A \rightarrow \epsilon$ (4) $B \rightarrow \epsilon$
- (B) (1) $S \rightarrow BbBa$ (2) $S \rightarrow AaAb$ (3) $A \rightarrow \epsilon$ (4) $B \rightarrow \epsilon$
- (C) (1) $S \rightarrow AaAb$ (2) $S \rightarrow BbBa$ (3) blank (4) blank
- (D) (1) $S \rightarrow BbBa$ (2) $S \rightarrow AaAb$ (3) blank (4) blank

Question 24 of 73 | GATE 2024 Set 2 (CS2 S6) | Q.41 (Computer Science & IT)

Q.41

Let M be the 5-state NFA with ϵ -transitions shown in the diagram below.

Which one of the following regular expressions represents the language accepted

by M ?

- (A) $\square(00)\square\square + \square 1(11)\square$
- (B) $\square 0\square + (1 + 0(00)\square)(11)\square\square$
- (C) $\square(00)\square + (1 + (00)\square)(11)\square\square$
- (D) $\square 0 + + 1(11)\square + 0(11)\square\square$

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Question 25 of 73 | GATE 2024 Set 2 (CS2 S6) | Q.52 (Computer Science & IT)

Q.52

Consider a context-free grammar G with the following 3 rules.

$S \rightarrow aS, S \rightarrow aSbS, S \rightarrow c$

Let $w \in L(G)$. Let $na(w)$, $nb(w)$, $nc(w)$ denote the number of times a , b , c occur in w , respectively. Which of the following statements is/are TRUE?

- (A) $na(w) > nb(w)$
- (B) $na(w) > nc(w) - 2$
- (C) $nc(w) = nb(w) + 1$
- (D) $nc(w) = nb(w) \square 2$

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Question 26 of 73 | GATE 2024 Set 2 (CS2 S6) | Q.62 (Computer Science & IT)

Q.62

Let L_1 be the language represented by the regular expression $b \square ab \square (ab \square ab \square) \square$

and $L_2 = \{\square w \square (a + b) \square \square \square | w| \leq 4\}$, where $|w|$ denotes the length of string w . The number of strings in L_2 which are also in L_1 is _____

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 27 of 73 | GATE 2024 Set 2 (CS2 S6) | Q.65 (Computer Science & IT)

Q.65

Consider the following augmented grammar, which is to be parsed with a SLR parser. The set of terminals is $\{a, b, c, d, \#, @\}$

$S' \rightarrow S$

$S \rightarrow SS \square \square Aa \square \square bAc \square \square Bc \square \square bBa$

$A \rightarrow d \#$

$B \rightarrow @$

Let $I_0 = \text{CLOSURE}(\{S' \rightarrow \square \bullet S\})$. The number of items in the set $\text{GOTO}(I_0, S)$ is _____

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 28 of 73 | GATE 2023 Single Session | Q.13 (Computer Science & IT)

Q.13

Consider the Deterministic Finite-state Automaton (DFA) A shown below. The DFA runs on the alphabet $\{0, 1\}$, and has the set of states $\{s, p, q, r\}$, with s being the start state and p being the only final state.

1
0
0
0
0,1
1
1
s
p
q
r

Which one of the following regular expressions correctly describes the language accepted by A?

- (A) $1(0 \cup 11)^*$
- (B) $0(0 + 1)^*$
- (C) $1(0 + 11)^*$
- (D) $1(110^*)^* \cup \epsilon$

Question 29 of 73 | GATE 2023 Single Session | Q.18 (Computer Science & IT)

Q.18

Consider the following definition of a lexical token id for an identifier in a programming language, using extended regular expressions:

letter

→

$[A-Za-z]$

digit

→

$[0-9]$

id

→

$letter (letter \mid digit)^*$

Which one of the following Non-deterministic Finite-state Automata with ϵ -transitions accepts the set of valid identifiers?

(A double-circle denotes a final state)

- (A) $letter \mid letter \mid digit \mid \epsilon$

- (B) letter letter digit ϵ ϵ
- (C) letter letter digit ϵ ϵ ϵ ϵ ϵ
- (D) letter letter digit ϵ CS

Question 30 of 73 | GATE 2023 Single Session | Q.23 (Computer Science & IT)

Q.23

Which of the following statements is/are CORRECT?

- (A) The intersection of two regular languages is regular.
- (B) The intersection of two context-free languages is context-free.
- (C) The intersection of two recursive languages is recursive.
- (D) The intersection of two recursively enumerable languages is recursively enumerable. CS

Question 31 of 73 | GATE 2023 Single Session | Q.38 (Computer Science & IT)

Q.38

Consider the context-free grammar G below

S
 $\rightarrow$
 $aSb \mid X$
 X
 $\rightarrow$
 $aX \mid Xb \mid a \mid b,$

where S and X are non-terminals, and a and b are terminal symbols. The starting non-terminal is S.

Which one of the following statements is CORRECT?

- (A) The language generated by G is $(a + b)^*$
- (B) The language generated by G is $a^* (a + b) b^*$
- (C) The language generated by G is $a^* b^* (a + b)$
- (D) The language generated by G is not a regular language CS

Question 32 of 73 | GATE 2023 Single Session | Q.39 (Computer Science & IT)

Q.39

Consider the pushdown automaton (PDA) P below, which runs on the input alphabet $\{a, b\}$, has stack alphabet $\{\square, A\}$, and has three states $\{s, p, q\}$, with s being the start state. A transition from state u to state v, labelled $c/X/\gamma$, where c is an input symbol or ϵ , X is a stack symbol, and γ is a string of stack symbols, represents the fact that in state u, the PDA can read c from the input, with X on the top of its stack, pop X from the stack, push in the string γ on the stack, and

go to state v . In the initial configuration, the stack has only the symbol $\square$ in it.
The PDA accepts by empty stack.

s
 $a/\square/A\square$
 $a/A/AA$
 p
 $b/A/\epsilon$
 $b/A/\epsilon$
 q
 $\epsilon/A/\epsilon$
 $\epsilon/\square/\epsilon$
 $\epsilon/A/\epsilon$
 $\epsilon/A/\epsilon$

Which one of the following options correctly describes the language accepted by P ?

- (A) $\{ambn \mid 1 \leq m \text{ and } n < m\}$
- (B) $\{ambn \mid 0 \leq n \leq m\}$
- (C) $\{ambn \mid 0 \leq m \text{ and } 0 \leq n\}$
- (D) $\{am \mid 0 \leq m\} \cup \{bn \mid 0 \leq n\}$ CS

Question 33 of 73 | GATE 2023 Single Session | Q.59 (Computer Science & IT)

Q.59

Consider the syntax directed translation given by the following grammar and semantic rules. Here N , I , F and B are non-terminals. N is the starting non-terminal, and $\#$, 0 and 1 are lexical tokens corresponding to input letters " $\#$ ", " 0 " and " 1 ", respectively. $X.val$ denotes the synthesized attribute (a numeric value) associated with a non-terminal X . $I1$ and $F1$ denote occurrences of I and F on the right hand side of a production, respectively. For the tokens 0 and 1 , $0.val = 0$ and $1.val = 1$.

N
 $\rightarrow$
 $I \# F$
 $N.val = I.val + F.val$
 I
 $\rightarrow$
 $I1 B$
 $I.val = (2 I1.val) + B.val$
 I
 $\rightarrow$
 B
 $I.val = B.val$
 F
 $\rightarrow$

B F1

F.val = 1

 $2(B.val + F1.val)$

F

→

B

F.val = 1

2B.val

B

→

0

B.val = 0.val

B

→

1

B.val = 1.val

The value computed by the translation scheme for the input string

10#011

is

. (Rounded off to three decimal places)

CS

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 34 of 73 | GATE 2023 Single Session | Q.62 (Computer Science & IT)

Q.62

CS

Consider the language L over the alphabet {0, 1}, given below:

 $L = \{w \in \{0, 1\}^* \mid w \text{ does not contain three or more consecutive 1's}\}.$

The minimum number of states in a Deterministic Finite-State Automaton (DFA)

for L is

.

CS

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 35 of 73 | GATE 2022 Single Session | Q.12 (Computer Science & IT)

Q.12

Which one of the following regular expressions correctly represents the language of the finite automaton given below?

(A) $0^*1^*0^*abababab0^*$ (B) $0^*1^*0^*(0^*1^*0^*)^*abababab0^*$

(C) * * * * * () () ab b ba a a b □ □

(D) * * * * * () () ba a ab b ab ba □ □

Question 36 of 73 | GATE 2022 Single Session | Q.13 (Computer Science & IT)

Q.13

Which one of the following statements is TRUE?

- (A) The (1) LALR parser for a grammar G cannot have reduce-reduce conflict if the (1) LR parser for G does not have reduce-reduce conflict.
- (B) Symbol table is accessed only during the lexical analysis phase.
- (C) Data flow analysis is necessary for run-time memory management.
- (D) (1) LR parsing is sufficient for deterministic context-free languages.

Question 37 of 73 | GATE 2022 Single Session | Q.23 (Computer Science & IT)

Q.23

Which of the following statements is/are TRUE?

- (A) Every subset of a recursively enumerable language is recursive.
- (B) If a language L and its complement $\bar{L}$ are both recursively enumerable, then L must be recursive.
- (C) Complement of a context-free language must be recursive.
- (D) If $1L$ and $2L$ are regular, then $1 \cup 2L$ must be deterministic context-free.

Question 38 of 73 | GATE 2022 Single Session | Q.29 (Computer Science & IT)

Q.29

Consider the augmented grammar with $\{ , , * , (,) ,$

$\}$

id

□

as the set of terminals.

S

S

□□

|

S

S

R R

□

□

*

|
R
R
P P
□
() |
P
S
id
□
If
O1 is the set of two
(0)
LR
items
•
•
{[
],[
]}
S
S
S
S
R
□□
□
□
, then
0
(
(),)
goto closure I
□ contains exactly _____ items.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 39 of 73 | GATE 2022 Single Session | Q.46 (Computer Science & IT)

Q.46

Which of the following is/are undecidable?

- (A) Given two Turing machines M_1 and M_2 , $M_1 M_2$ decide if $L_1 \cap L_2 = \emptyset$.
- (B) Given a Turing machine M , decide if $L(M)$ is regular.

- (C) Given a Turing machine , M decide if M accepts all strings.
- (D) Given a Turing machine , M decide if M takes more than 1073 steps on every string.

Question 40 of 73 | GATE 2022 Single Session | Q.47 (Computer Science & IT)

Q.47

Consider the following languages:

 $*$
 1
 $\{$
 $|$
 $\{, \}$
 n
 n
 L
 $a wa$
 w
 $a b$
 $\square$
 $\square$
 $*$
 2
 $\{$
 $|$
 $,$
 $\{, \}, |$
 $|, |$
 $| 0 \}$
 R
 L
 $w x w$
 $w x$
 $a b$
 w
 x
 $\square$
 $\square$
 $\square$

Note that

 R
 w is the reversal of the string .

 w Which of the following is/are TRUE?

- (A) 1L and 2L are regular.
- (B) 1L and 2L are context-free.
- (C) 1L is regular and 2L is context-free.
- (D) 1L and 2L are context-free but not regular.

Question 41 of 73 | GATE 2022 Single Session | Q.48 (Computer Science & IT)

Q.48

Consider the following languages:

 $*$

1

{

|

{, }

L

ww w

a b

□

□

2

{

|

,

0}

n

n

m

L

a b c

m n

□

□

3

{

|

,

0}

m

n

n

L

a b c

m n

□

□

Which of the following statements is/are FALSE?

- (A) 1L is not context-free but 2L and 3L are deterministic context-free.
- (B) Neither 1L nor 2L is context-free.
- (C) 2L, 3L and $2^3 L L^2$ all are context-free.
- (D) Neither 1L nor its complement is context-free.

Question 42 of 73 | GATE 2022 Single Session | Q.65 (Computer Science & IT)

Q.65

Consider the following grammar along with translation rules.

1

1

•

•

•

#

{

*

}

val

val

val

S

S

T

S

S

T

□

□

•

•

{

}

val

val

S

T

S

T
□
□
1
•
1•
•
%
{
}
val
val
val
T
T
R
T
T
R
□
□
□
•
•
{
}
val
val
T
R
T
R
□
□
•
•
{
}
val
val
R
id
R

id

□

□

Here # and % are operators and id is a token that represents an integer and

•val

id

represents the corresponding integer value. The set of non-terminals is { , ,

, }

S T R P

and a subscripted non-terminal indicates an instance of the non-terminal.

Using this translation scheme, the computed value of

•val

S

for root of the parse tree

for the expression $20\#10\%5\#8\%2\%2$ is _____.

GATE 2022

Part 2 of 2

official answer key (not a question paper) | 2 pages

Paper 7 of 16

Q. No.

Session

Question

Type

Subject

Name

Key/Range

Mark

1

1

MCQ

GA

D

1

2

1

MCQ

GA

C

1

3

1

MCQ

GA

D
1
4
1
MCQ
GA
A
1
5
1
MCQ
GA
B
1
6
1
MCQ
GA
B
2
7
1
MCQ
GA
D
2
8
1
MCQ
GA
D
2
9
1
MCQ
GA
B
2
10
1
MCQ
GA
D

2
11
1
MCQ
CS
A
1
12
1
MCQ
CS
D
1
13
1
MCQ
CS
D
1
14
1
MCQ
CS
A
1
15
1
MCQ
CS
A
1
16
1
MCQ
CS
B
1
17
1
MCQ
CS
A
1

18
1
MCQ
CS
B
1
19
1
MCQ
CS
C
1
20
1
MCQ
CS
C
1
21
1
MCQ
CS
D
1
22
1
MCQ
CS
C
1
23
1
MSQ
CS
B,C,D
1
24
1
MSQ
CS
A,B,D
1
25

1
MSQ
CS
A,B
1
26
1
MSQ
CS
A,D
1
27
1
MSQ
CS
A,B,C
1
28
1
NAT
CS
509 to 509
1
29
1
NAT
CS
5 to 5
1
30
1
NAT
CS
36 to 36
1
31
1
NAT
CS
8 to 8
1
32
1

NAT

CS

7 to 7

1

33

1

NAT

CS

0.85 to 0.85

1

34

1

NAT

CS

-0.5 to -0.5

1

35

1

NAT

CS

4 to 4

1

36

1

MCQ

CS

A

2

37

1

MCQ

CS

D

2

38

1

MCQ

CS

C

2

39

1

MCQ

CS
A
2
40
1
MCQ
CS
C
2
41
1
MCQ
CS
B
2
42
1
MCQ
CS
D
2
43
1
MCQ
CS
A
2
44
1
MCQ
CS
A
2
45
1
MCQ
CS
D
2
46
1
MSQ
CS

A,B,C
2
47
1
MSQ
CS
A,B,C
2
48
1
MSQ
CS
B,C,D
2
49
1
MSQ
CS
A,B,C
2
50
1
MSQ
CS
A,B,C
2
51
1
MSQ
CS
A,B,C
2
52
1
MSQ
CS
A
2
53
1
MSQ
CS
A,C,D

2
54
1
MSQ
CS
A,B,D
2
55
1
MSQ
CS
B,D
2
56
1
NAT
CS
2 to 2
2
57
1
NAT
CS
0.5 to 0.5
2
58
1
NAT
CS
24 to 24
2
59
1
NAT
CS
7.07 to 7.09
2
60
1
NAT
CS
33 to 33
2

61
1
NAT
CS
1.42 to 1.45
2
62
1
NAT
CS
0 to 0
2
63
1
NAT
CS
153 to 153
2
64
1
NAT
CS
0.6 to 0.6
2
65
1
NAT
CS
80 to 80
2

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 43 of 73 | GATE 2021 Set 1 (CS1) | Q.5 (Computer Science & IT)

Q.5

_____ is to surgery as writer is to _____

Which one of the following options maintains a similar logical relation in the above sentence?

- (A) Plan, outline
- (B) Hospital, library
- (C) Doctor, book

- (D) Medicine, grammar Q. 6 – Q. 10 Multiple Choice Question (MCQ), carry TWO marks each (for each wrong answer: – 2/3).

Question 44 of 73 | GATE 2019 Single Session | Q.7 (Computer Science & IT)

Q.7

If L is a regular language over $\Sigma = \{a, b\}$, which one of the following languages is NOT regular ?

- (A) $L \cup LR = \{xy \mid x \in L, y \in R\}$
- (B) $\{ww^R \mid w \in L\}$
- (C) $\text{Prefix}(L) = \{x \in \Sigma^* \mid \exists y \in \Sigma^* \text{ such that } xy \in L\}$
- (D) $\text{Suffix}(L) = \{y \in \Sigma^* \mid \exists x \in \Sigma^* \text{ such that } xy \in L\}$

Question 45 of 73 | GATE 2019 Single Session | Q.15 (Computer Science & IT)

Q.15 For $\Sigma = \{a, b\}$, let us consider the regular language $L = \{x \mid x = a^{2+3k} \text{ or } x = b^{10+12k}, k \geq 0\}$. Which one of the following can be a pumping length (the constant guaranteed by the pumping lemma) for L ?

- (A) 3
- (B) 5
- (C) 9
- (D) 24

Question 46 of 73 | GATE 2019 Single Session | Q.19 (Computer Science & IT)

Q.19 Consider the grammar given below:

$S \rightarrow Aa$

$A \rightarrow BD$

$B \rightarrow b \mid \epsilon$

$D \rightarrow d \mid \epsilon$

Let a, b, d , and $\$$ be indexed as follows:

a
 b
 d
 $\$$
 3
 2
 1
 0

Compute the FOLLOW set of the non-terminal B and write the index values for the symbols

in the FOLLOW set in the descending order. (For example, if the FOLLOW set is {a, b, d, \$}, then the answer should be 3210)

Answer: _____.

CS

5/15

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 47 of 73 | GATE 2019 Single Session | Q.31 (Computer Science & IT)

Q.31 Which one of the following languages over $\Sigma = \{a, b\}$ is NOT context-free?

- (A) $\{wwR \mid w \in \{a, b\}^*\}$
- (B) $\{wanbnwR \mid w \in \{a, b\}^*, n \geq 0\}$
- (C) $\{wanwRbn \mid w \in \{a, b\}^*, n \geq 0\}$
- (D) $\{anbi \mid i \in \{n, 3n, 5n\}, n \geq 0\}$ CS 8/15

Question 48 of 73 | GATE 2019 Single Session | Q.34 (Computer Science & IT)

Q.34 Consider the following sets:

S1.

Set of all recursively enumerable languages over the alphabet $\{0,1\}$

S2.

Set of all syntactically valid C programs

S3.

Set of all languages over the alphabet $\{0,1\}$

S4.

Set of all non-regular languages over the alphabet $\{0,1\}$

Which of the above sets are uncountable?

- (A) S1 and S2
- (B) S3 and S4
- (C) S2 and S3
- (D) S1 and S4

Question 49 of 73 | GATE 2019 Single Session | Q.36 (Computer Science & IT)

Q.36 Consider the following grammar and the semantic actions to support the inherited type declaration attributes. Let X_1, X_2, X_3, X_4, X_5 , and X_6 be the placeholders for the non-terminals D, T, L or L_1 in the following table:

Production rule

Semantic action

$D \rightarrow T L$

$X1.type = X2.type$

$T \rightarrow int$

$T.type = int$

$T \rightarrow float$

$T.type = float$

$L \rightarrow L1, id$

$X3.type = X4.type$

$addType(id.entry, X5.type)$

$L \rightarrow id$

$addType(id.entry, X6.type)$

Which one of the following are the appropriate choices for $X1$, $X2$, $X3$ and $X4$?

- (A) $X1 = L, X2 = T, X3 = L1, X4 = L$
- (B) $X1 = T, X2 = L, X3 = L1, X4 = T$
- (C) $X1 = L, X2 = L, X3 = L1, X4 = T$
- (D) $X1 = T, X2 = L, X3 = T, X4 = L1$

Question 50 of 73 | GATE 2019 Single Session | Q.43 (Computer Science & IT)

Q.43 Consider the augmented grammar given below:

$S' \rightarrow S$

$S \rightarrow \square L \square \mid id$

$L \rightarrow L, S \mid S$

Let $I0 = CLOSURE(\{[S' \rightarrow \bullet S]\})$. The number of items in the set $GOTO(I0, \square)$ is:

_____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 51 of 73 | GATE 2019 Single Session | Q.48 (Computer Science & IT)

Q.48 Let Σ be the set of all bijections from $\{1, \dots, 5\}$ to $\{1, \dots, 5\}$, where id denotes the identity function, i.e. $id(j) = j, \square j$. Let $\square$ denote composition on functions. For a string $x =$

$x_1 x_2 \dots x_n \square \Sigma^n, n \geq 0$, let $\pi(x) = x_1 \square x_2 \square \dots \square x_n$.

Consider the language $L = \{x \square \Sigma^n \mid \pi(x) = id\}$. The minimum number of states in any DFA accepting L is .

CS

12/15

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 52 of 73 | GATE 2018 Single Session | Q.6 (Computer Science & IT)

Q.6

Let N be an NFA with n states. Let k be the number of states of a minimal DFA which is equivalent to N . Which one of the following is necessarily true?

- (A) $k \geq 2n$
- (B) $k \geq n$
- (C) $k \leq n^2$
- (D) $k \leq 2n$

Question 53 of 73 | GATE 2018 Single Session | Q.7 (Computer Science & IT)

Q.7

The set of all recursively enumerable languages is

- (A) closed under complementation.
- (B) closed under intersection.
- (C) a subset of the set of all recursive languages.
- (D) an uncountable set.

Question 54 of 73 | GATE 2018 Single Session | Q.8 (Computer Science & IT)

Q.8

Which one of the following statements is FALSE?

- (A) Context-free grammar can be used to specify both lexical and syntax rules.
- (B) Type checking is done before parsing.
- (C) High-level language programs can be translated to different Intermediate Representations.
- (D) Arguments to a function can be passed using the program stack. CS 3/20

Question 55 of 73 | GATE 2018 Single Session | Q.35 (Computer Science & IT)

Q.35 Consider the following languages:

- I. $\{ambncpdq \mid m + p = n + q, \text{ where } m, n, p, q \geq 0\}$
- II. $\{ambncpdq \mid m = n \text{ and } p = q, \text{ where } m, n, p, q \geq 0\}$
- III. $\{ambncpdq \mid m = n = p \text{ and } p \neq q, \text{ where } m, n, p, q \geq 0\}$
- IV. $\{ambncpdq \mid mn = p + q, \text{ where } m, n, p, q \geq 0\}$

Which of the languages above are context-free?

- (A) I and IV only
- (B) I and II only
- (C) II and III only
- (D) II and IV only

Question 56 of 73 | GATE 2018 Single Session | Q.36 (Computer Science & IT)

Q.36 Consider the following problems. $L(G)$ denotes the language generated by a grammar G .

$L(M)$ denotes the language accepted by a machine M .

(I)

For an unrestricted grammar G and a string w , whether $w \in L(G)$

(II)

Given a Turing machine M , whether $L(M)$ is regular

(III)

Given two grammars G_1 and G_2 , whether $L(G_1) = L(G_2)$

(IV)

Given an NFA N , whether there is a deterministic PDA P such that N and P accept the same language.

Which one of the following statements is correct?

- (A) Only I and II are undecidable
- (B) Only III is undecidable
- (C) Only II and IV are undecidable
- (D) Only I, II and III are undecidable CS 12/20

Question 57 of 73 | GATE 2018 Single Session | Q.37 (Computer Science & IT)

Q.37 A lexical analyzer uses the following patterns to recognize three tokens T_1 , T_2 , and T_3 over the alphabet $\{a, b, c\}$.

T_1 : $a? (b|c)\square a$

T_2 : $b? (a|c)\square b$

T_3 : $c? (b|a)\square c$

Note that ' $x?$ ' means 0 or 1 occurrence of the symbol x . Note also that the analyzer outputs the token that matches the longest possible prefix.

If the string $bbaacabc$ is processed by the analyzer, which one of the following is the sequence of tokens it outputs?

- (A) $T_1T_2T_3$
- (B) $T_1T_1T_3$
- (C) $T_2T_1T_3$
- (D) T_3T_3

Question 58 of 73 | GATE 2018 Single Session | Q.52 (Computer Science & IT)

Q.52 Given a language L , define L_i as follows:

$L_0 = \{\epsilon\}$

$L_i = L_{i-1} \square L$ for all $i > 0$

The order of a language L is defined as the smallest k such that $L_k = L_{k+1}$.

Consider the language L_1 (over alphabet $\{0\}$) accepted by the following automaton.

The order of L1 is ____.

0
0
0
CS
19/20

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 59 of 73 | GATE 2016 Set 1 (CS1) | Q.16 (Computer Science & IT)

Q.16

Which of the following languages is generated by the given grammar?

S
→
aS | bS | ε

- (A) $\{anbm \mid n, m \geq 0\}$
- (B) $\{w \in \{a,b\}^* \mid w \text{ has equal number of a's and b's}\}$
- (C) $\{an \mid n \geq 0\} \cup \{bn \mid n \geq 0\} \cup \{anbn \mid n \geq 0\}$
- (D) $\{a,b\}^*$ CS(Set A) 5/17 GATE 2016

Question 60 of 73 | GATE 2016 Set 1 (CS1) | Q.17 (Computer Science & IT)

Q.17

Which of the following decision problems are undecidable?

- I. Given NFAs N_1 and N_2 , is $L(N_1) \cap L(N_2) = \Phi$?
- II. Given a CFG $G = (N, \Sigma, P, S)$ and a string $x \in \Sigma^*$, does $x \in L(G)$?
- III. Given CFGs G_1 and G_2 , is $L(G_1) = L(G_2)$?
- IV. Given a TM M , is $L(M) = \Phi$?

- (A) I and IV only
- (B) II and III only
- (C) III and IV only
- (D) II and IV only

Question 61 of 73 | GATE 2016 Set 1 (CS1) | Q.18 (Computer Science & IT)

Q.18

Which one of the following regular expressions represents the language: the set of all binary strings having two consecutive 0s and two consecutive 1s?

- (A) $(0+1)^*0011(0+1)^* + (0+1)^*1100(0+1)^*$

- (B) $(0+1) \square (00(0+1) \square 11+11(0+1) \square 00)(0+1) \square$
- (C) $(0+1) \square 00(0+1) \square + (0+1) \square 11(0+1) \square$
- (D) $00(0+1) \square 11+11(0+1) \square 00$

Question 62 of 73 | GATE 2016 Set 1 (CS1) | Q.42 (Computer Science & IT)

Q.42

Consider the following context-free grammars:

G1: $S \rightarrow aS|B, B \rightarrow b|bB$

G2: $S \rightarrow aA|bB, A \rightarrow aA|B|\epsilon, B \rightarrow bB|\epsilon$

Which one of the following pairs of languages is generated by G1 and G2, respectively?

- (A) $\{ambn|m > 0 \text{ or } n > 0\}$ and $\{ambn|m > 0 \text{ and } n > 0\}$
- (B) $\{ambn|m > 0 \text{ and } n > 0\}$ and $\{ambn|m > 0 \text{ or } n \geq 0\}$
- (C) $\{ambn|m \geq 0 \text{ or } n > 0\}$ and $\{ambn|m > 0 \text{ and } n > 0\}$
- (D) $\{ambn|m \geq 0 \text{ and } n > 0\}$ and $\{ambn|m > 0 \text{ or } n > 0\}$ CS(Set A) 13/17 GATE 2016

Question 63 of 73 | GATE 2016 Set 1 (CS1) | Q.43 (Computer Science & IT)

Q.43

Consider the transition diagram of a PDA given below with input alphabet $\Sigma = \{a,b\}$ and stack alphabet $\Gamma = \{X,Z\}$. Z is the initial stack symbol. Let L denote the language accepted by the PDA.

$b, X/\epsilon$

$\epsilon, Z/Z$

$b, X/\epsilon$

$a, Z/XZ$

$a, X/XX$

Which one of the following is TRUE?

- (A) $L = \{anbn|n \geq 0\}$ and is not accepted by any finite automata
- (B) $L = \{an|n \geq 0\} \square \{anbn|n \geq 0\}$ and is not accepted by any deterministic PDA
- (C) L is not accepted by any Turing machine that halts on every input
- (D) $L = \{an|n \geq 0\} \square \{anbn|n \geq 0\}$ and is deterministic context-free

Question 64 of 73 | GATE 2016 Set 1 (CS1) | Q.44 (Computer Science & IT)

Q.44

Let X be a recursive language and Y be a recursively enumerable but not recursive language.

Let W and Z be two languages such that Y reduces to W, and Z reduces to X (reduction means the standard many-one reduction). Which one of the following statements is TRUE?

- (A) W can be recursively enumerable and Z is recursive.
- (B) W can be recursive and Z is recursively enumerable.
- (C) W is not recursively enumerable and Z is recursive.
- (D) W is not recursively enumerable and Z is not recursive.

Question 65 of 73 | GATE 2016 Set 2 (CS2) | Q.16 (Computer Science & IT)

Q.16

The number of states in the minimum sized DFA that accepts the language defined by the regular expression

$$(0+1)^*(0+1)(0+1)^*$$

is

.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 66 of 73 | GATE 2016 Set 2 (CS2) | Q.17 (Computer Science & IT)

Q.17

Language L1 is defined by the grammar: $S_1 \rightarrow aS_1b | \epsilon$

Language L2 is defined by the grammar: $S_2 \rightarrow abS_2 | \epsilon$

Consider the following statements:

P: L1 is regular

Q: L2 is regular

Which one of the following is TRUE?

- (A) Both P and Q are true
- (B) P is true and Q is false
- (C) P is false and Q is true
- (D) Both P and Q are false CS(Set B) 4/18 GATE 2016

Question 67 of 73 | GATE 2016 Set 2 (CS2) | Q.18 (Computer Science & IT)

Q.18

Consider the following types of languages: L1 : Regular, L2 : Context-free, L3 : Recursive, L4 : Recursively enumerable. Which of the following is/are TRUE?

I. $L_3 \cap L_4$ is recursively enumerable

II. $L_2 \cap L_3$ is recursive

III. L_1

$\cap L_2$ is context-free

IV. $L_1 \cap L_2$ is context-free

- (A) I only

- (B) I and III only
- (C) I and IV only
- (D) I, II and III only

Question 68 of 73 | GATE 2016 Set 2 (CS2) | Q.19 (Computer Science & IT)

Q.19

Match the following:

- (P) Lexical analysis
 (i) Leftmost derivation
 (Q) Top down parsing
 (ii) Type checking
 (R) Semantic analysis
 (iii) Regular expressions
 (S) Runtime environments
 (iv) Activation records

- (A) $P \leftrightarrow i, Q \leftrightarrow ii, R \leftrightarrow iv, S \leftrightarrow iii$
- (B) $P \leftrightarrow iii, Q \leftrightarrow i, R \leftrightarrow ii, S \leftrightarrow iv$
- (C) $P \leftrightarrow ii, Q \leftrightarrow iii, R \leftrightarrow i, S \leftrightarrow iv$
- (D) $P \leftrightarrow iv, Q \leftrightarrow i, R \leftrightarrow ii, S \leftrightarrow iii$

Question 69 of 73 | GATE 2016 Set 2 (CS2) | Q.42 (Computer Science & IT)

Q.42

Consider the following two statements:

- I. If all states of an NFA are accepting states then the language accepted by the NFA is Σ^* .
 II. There exists a regular language A such that for all languages B, $A \cap B$ is regular.

Which one of the following is CORRECT?

- (A) Only I is true
- (B) Only II is true
- (C) Both I and II are true
- (D) Both I and II are false

Question 70 of 73 | GATE 2016 Set 2 (CS2) | Q.43 (Computer Science & IT)

Q.43

Consider the following languages:

$$L1 = \{anbmcm+n : m, n \geq 1\}$$

$$L2 = \{anbnc2n : n \geq 1\}$$

Which one of the following is TRUE?

- (A) Both L1 and L2 are context-free.
- (B) L1 is context-free while L2 is not context-free.
- (C) L2 is context-free while L1 is not context-free.
- (D) Neither L1 nor L2 is context-free. CS(Set B) 12/18 GATE 2016

Question 71 of 73 | GATE 2016 Set 2 (CS2) | Q.44 (Computer Science & IT)

Q.44

Consider the following languages.

L1

=

 $\{\square M \square \mid M \text{ takes at least 2016 steps on some input}\}$,

L2

=

 $\{\square M \square \mid M \text{ takes at least 2016 steps on all inputs}\}$ and

L3

=

 $\{\square M \square \mid M \text{ accepts } \epsilon\}$,

where for each Turing machine M , $\square M \square$ denotes a specific encoding of M . Which one of the following is TRUE?

- (A) L1 is recursive and L2, L3 are not recursive
- (B) L2 is recursive and L1, L3 are not recursive
- (C) L1, L2 are recursive and L3 is not recursive
- (D) L1, L2, L3 are recursive

Question 72 of 73 | GATE 2016 Set 2 (CS2) | Q.45 (Computer Science & IT)

Q.45

Which one of the following grammars is free from left recursion?

- (A) $S \rightarrow AB \mid Aa \mid b \mid B \rightarrow c$
- (B) $S \rightarrow Ab \mid Bb \mid c \mid A \rightarrow Bd \mid \epsilon \mid B \rightarrow e$
- (C) $S \rightarrow Aa \mid BA \rightarrow Bb \mid Sc \mid \epsilon \mid B \rightarrow d$
- (D) $S \rightarrow Aa \mid Bb \mid c \mid A \rightarrow Bd \mid \epsilon \mid B \rightarrow Ae \mid \epsilon$ CS(Set B) 13/18 GATE 2016

Question 73 of 73 | GATE 2016 Set 2 (CS2) | Q.46 (Computer Science & IT)

Q.46

A student wrote two context-free grammars G_1 and G_2 for generating a single C-like array declaration. The dimension of the array is at least one. For example,

```
int a[10][3];
```

The grammars use D as the start symbol, and use six terminal symbols int ; id [] num.

Grammar G1

Grammar G2

$D \rightarrow \text{int}L;$

$D \rightarrow \text{int}L;$

$L \rightarrow \text{id}[E$

$L \rightarrow \text{id}E$

$E \rightarrow \text{num}]$

$E \rightarrow E[\text{num}]$

$E \rightarrow \text{num}][E$

$E \rightarrow [\text{num}]$

Which of the grammars correctly generate the declaration mentioned above?

- (A) Both G1 and G2
- (B) Only G1
- (C) Only G2
- (D) Neither G1 nor G2